

STM32F103C8T6 Custom Development Board

Project Overview

Designed a custom **STM32F103C8T6-based PCB** using KiCad, covering the complete hardware design workflow from schematic capture and component selection to PCB layout, routing, verification, and 3D visualization.

Key Features

- **STM32F103C8T6** 32-bit microcontroller
- USB Micro-B interface
- 3.3 V regulated power supply using **AMS1117-3.3**
- **SWD** programming and debugging interface
- **UART** communication header
- **I²C** expansion interface
- BOOT and reset controls
- 8 MHz external crystal oscillator
- On-board power/status LED
- Four mounting holes

PCB Design

- Schematic capture and circuit organization
- Component and footprint selection
- 2-layer PCB layout and routing
- SMD and through-hole component integration
- Board-edge connector placement
- Silkscreen labeling for easier interface identification
- PCB design verification using ERC/DRC
- 3D board visualization

Tools & Technologies

KiCad | STM32F103C8T6 | PCB Design | Schematic Capture | PCB Layout

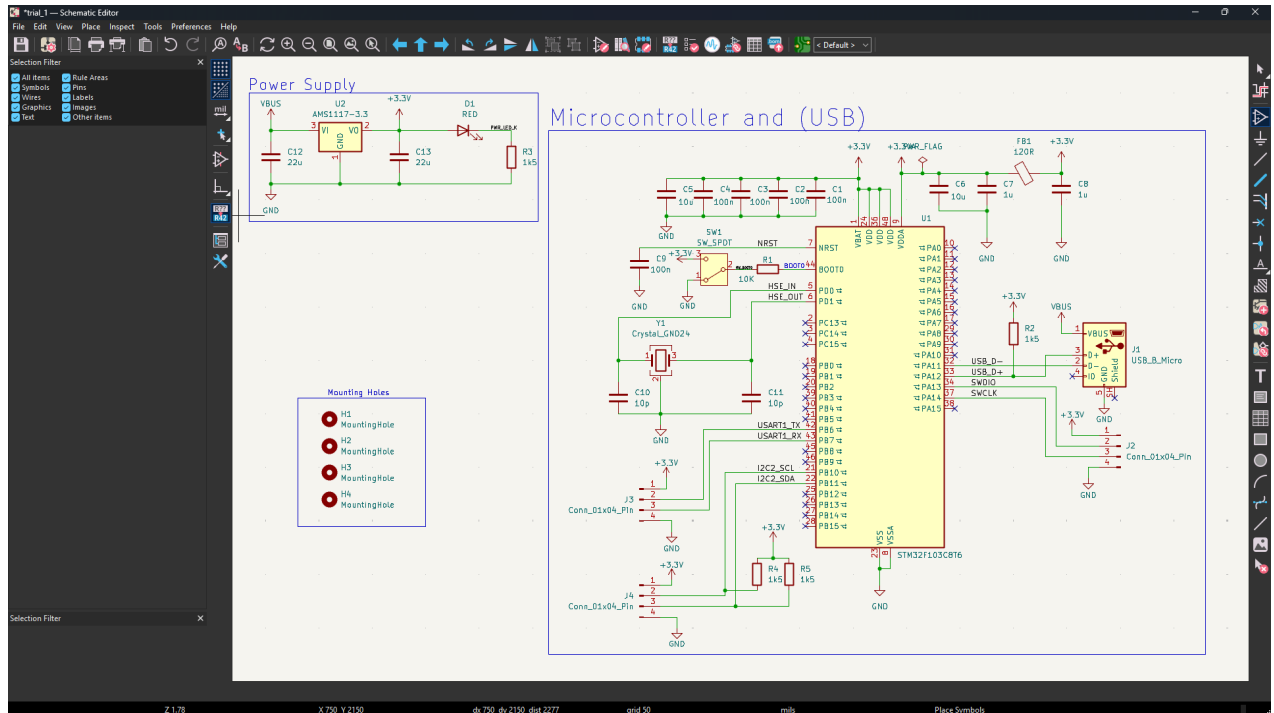
Deliverables

Schematic | PCB Layout | 3D Model

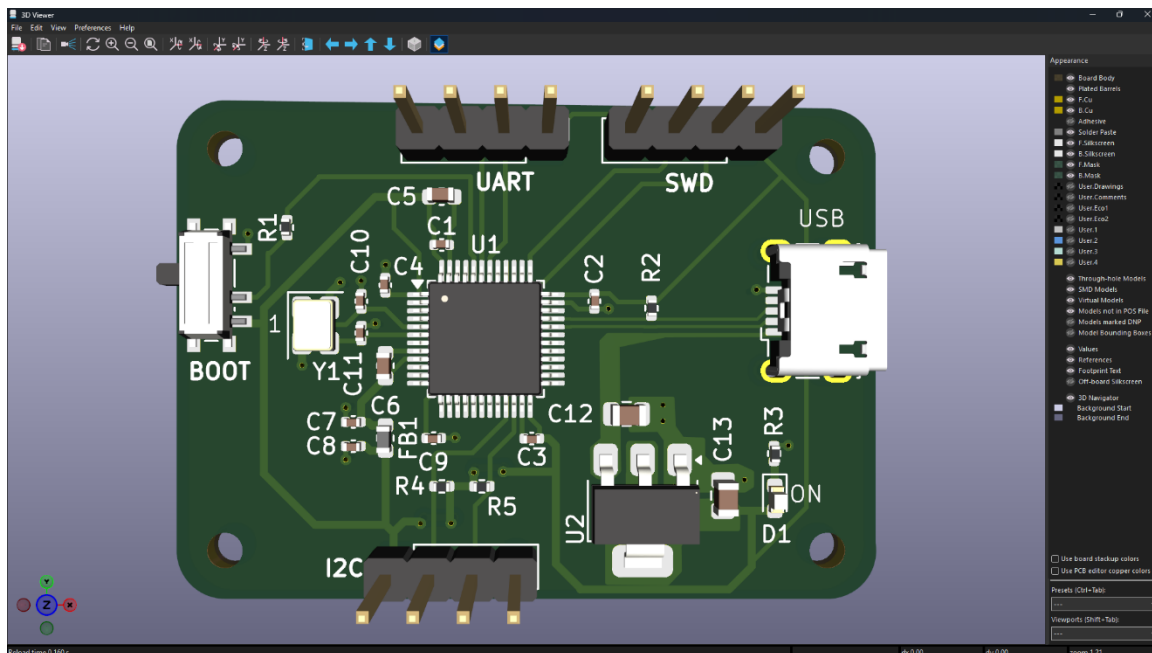
STM32F103C8T6 Custom Development Board

A custom STM32-based PCB designed from schematic to 3D visualization.

Schematic Design



3D Visualization



STM32WB55 Wireless & RF PCB Design

Project Overview

Designed a custom **STM32WB55CEU6-based wireless PCB** using KiCad, covering schematic design, component selection, PCB layout, routing, and 3D visualization.

Key Features

- **STM32WB55CEU6** wireless microcontroller
- USB-C interface
- 5 V to 3.3 V regulated power supply using **MIC5365-3.3**
- SWD programming and debugging interface using Tag-Connect
- UART communication header
- BOOT0 control switch
- **32 MHz HSE** and **32.768 kHz LSE** crystal circuits
- RF matching network
- **u.FL RF connector** for RF connectivity
- Fiducial markers and four mounting holes

PCB Design

- Complete schematic capture and component integration
- SMD component placement and footprint selection
- 2-layer PCB layout and routing
- Power and signal routing
- Ground/copper-plane implementation
- RF section layout with matching components and u.FL interface
- PCB design verification
- 3D board visualization

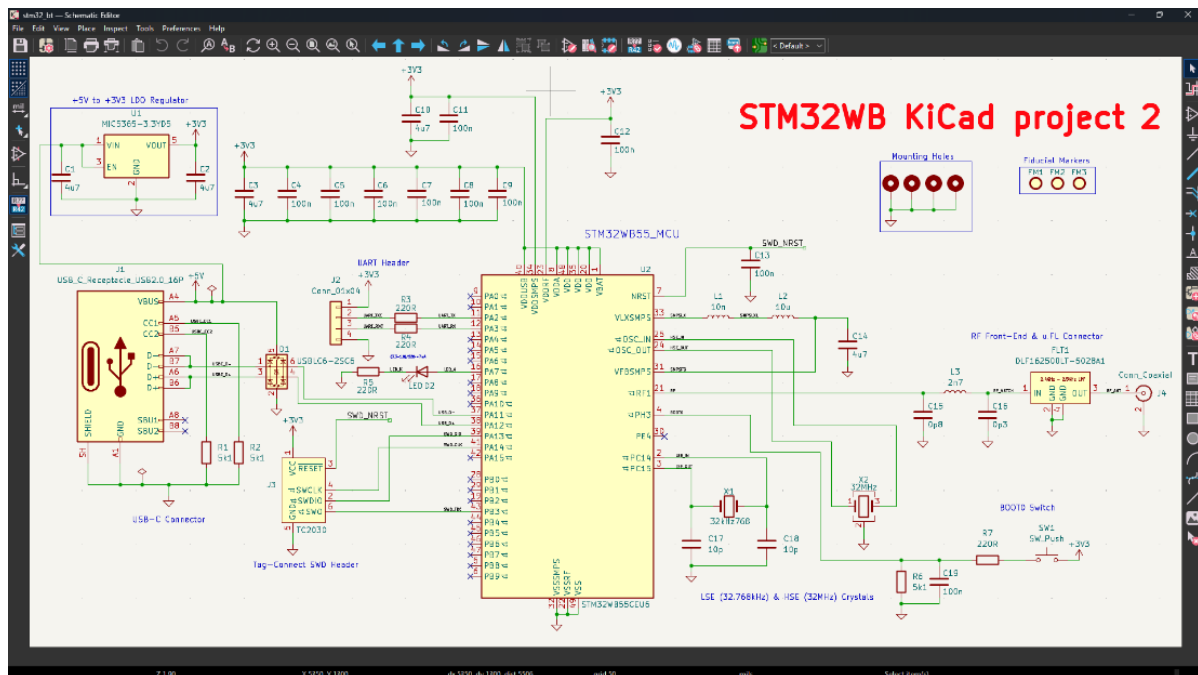
Tools & Technologies

KiCad | STM32WB55CEU6 | PCB Design | Schematic Capture | PCB Layout | RF Layout | USB-C | SWD

Deliverables

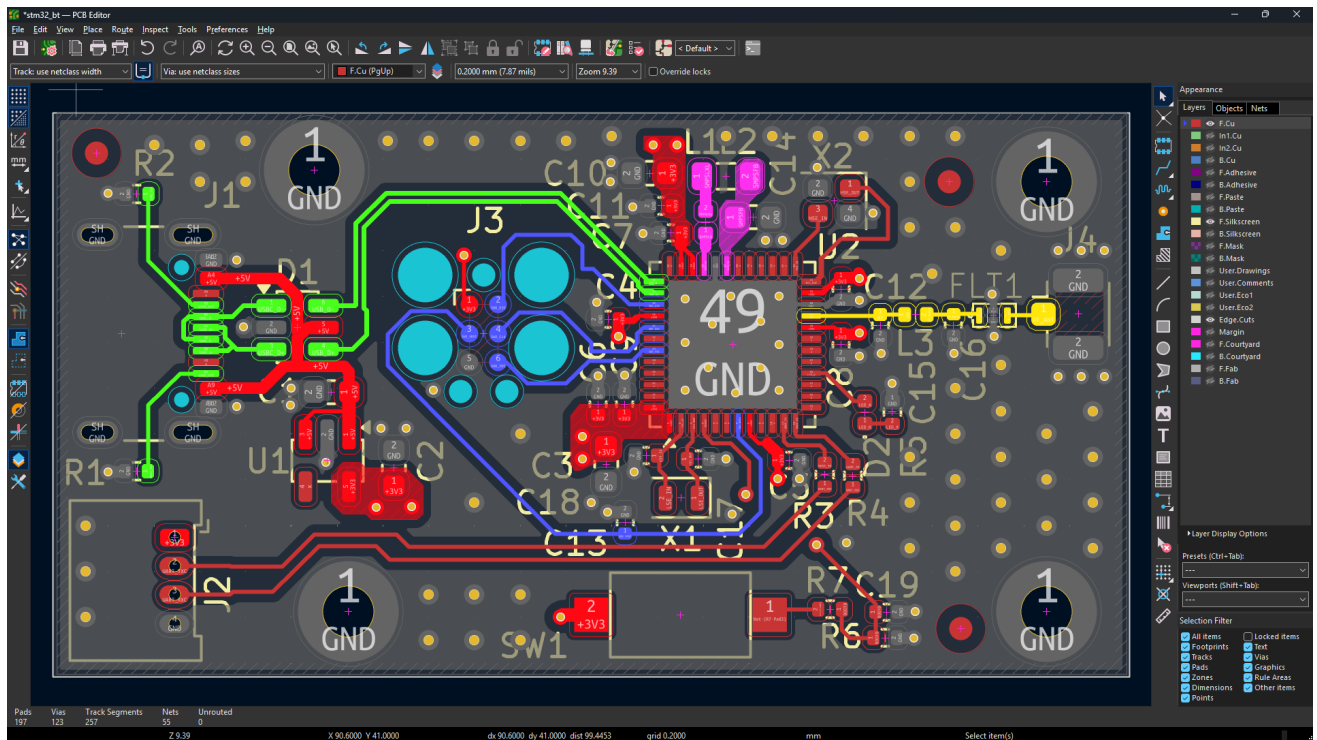
Schematic | PCB Layout | Layer Views | 3D Model

Schematic Design

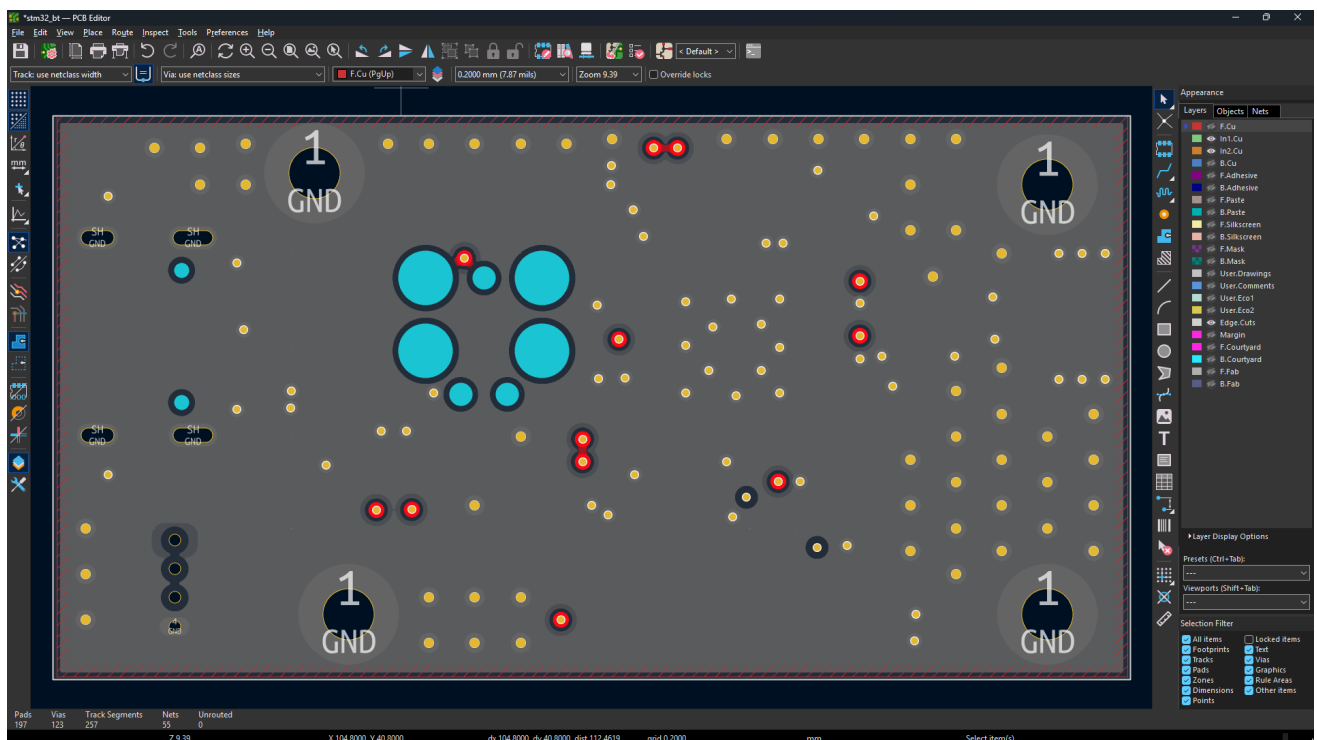


PCB Layout

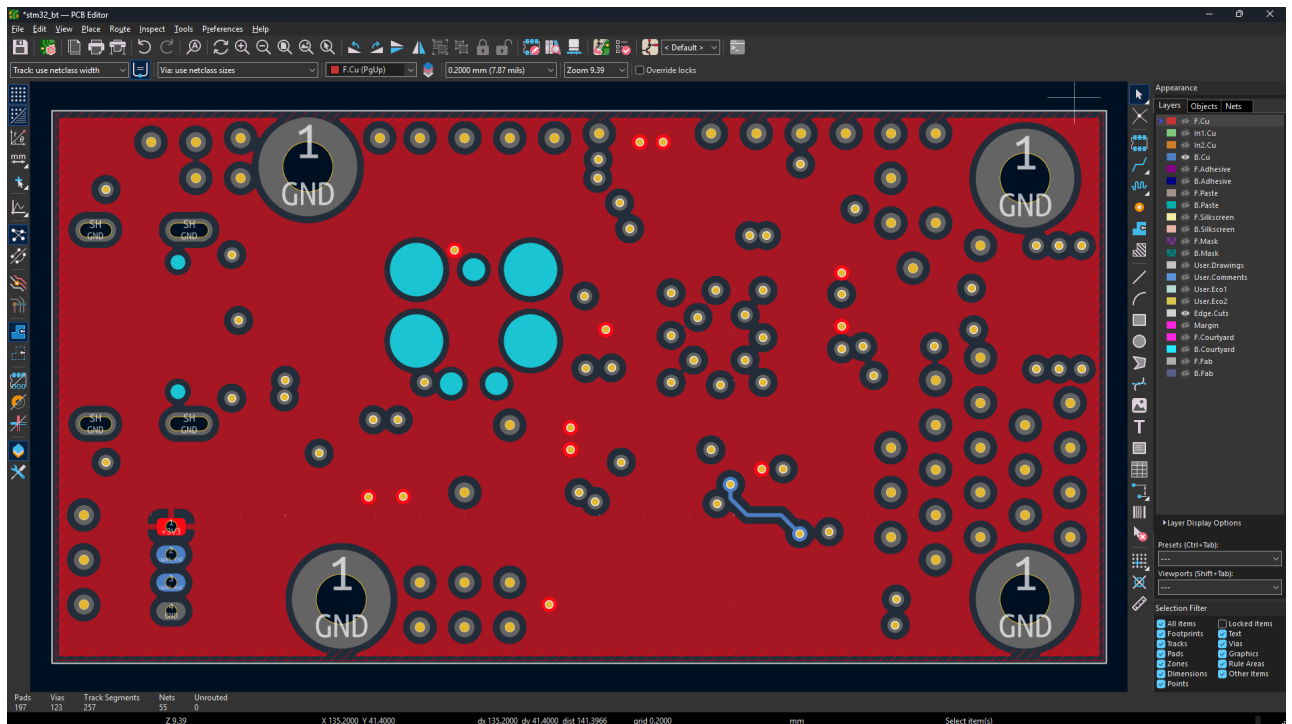
Layer 1:



Layer 2 and 3:



Layer 4:



3D Visualization

